An exact evaluation of the two-shifted q-Motzkin determinant

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Abstract

We evaluate the determinant for every size, using determinant condensation and Cigler's established zero- and one-shift evaluations. This determines the missing power of q and the complete normalized polynomial. The printed conjecture omits a constant term when the size is divisible by three. Campaign eligibility remains pending: the complete final journal source has not been inspected. No new research credit is claimed.

Task 1: the original problem and its complete evaluation

Let $M_0(q) = 1$ and

$$M_{n+1}(q) = M_n(q) + \sum_{r=0}^{n-1} q^{r+1} M_r(q) M_{n-r-1}(q).$$

Put

$$D_n^{(s)} = \det[M_{i+j+s}(q)]_{i,j=0}^{n-1}, \qquad D_0^{(s)} = 1, \qquad S_n = \sum_{j=0}^{n-1} j^2 = \frac{n(n-1)(2n-1)}{6}.$$

MathDB 339076 reproduces Conjecture A.2 on p. 62 of [2]. It predicts $D_n^{(2)} = q^{c_n} f_n(q)$ with

$$f_n(q) = \begin{cases} \displaystyle\sum_{\substack{1 \le l \le n \\ l \not\equiv 1 \pmod 3}} q^l, & n \equiv 0 \pmod 3, \\ \displaystyle(1+q) \sum_{r=0}^{\lfloor n/3 \rfloor} q^{3r}, & n \not\equiv 0 \pmod 3. \end{cases}$$

The lower bound $1 \le l$ in the first branch is also present in the primary preprint. The following theorem evaluates the whole family, not merely the first exception to that printed formula.

Theorem 1. *For every integer $n \ge 0$, the following polynomial identity holds:*

$$D_n^{(2)} = q^{S_n + n^2} \sum_{\substack{0 \le j \le n \\ j \not\equiv 2 \pmod 3}} q^{-j}. \tag{1}$$

For $n \geq 1$, equivalently $D_n^{(2)} = q^{c_n} g_n(q)$, where

$$c_n = S_n + n^2 - n + \begin{cases} 1, & n \equiv 2 \pmod 3, \\ 0, & \textit{otherwise}, \end{cases} \tag{2}$$

$$g_n(q) = \begin{cases} \displaystyle\sum_{\substack{0 \leq l \leq n \\ l \not\equiv 1 \pmod 3}} q^l, & n \equiv 0 \pmod 3, \\ \displaystyle(1+q) \sum_{r=0}^{\lfloor n/3 \rfloor} q^{3r}, & n \not\equiv 0 \pmod 3. \end{cases} \tag{3}$$

In particular, every nonzero coefficient of g_n is 1; $g_n(0) = 1$; and c_n is the exact order of vanishing at $q = 0$.

A condensation proof

Cigler's established formulas [1, Eqs. (39)–(40)] are

$$D_n^{(0)} = q^{S_n}, \qquad D_n^{(1)} = \kappa_n q^{S_n + n(n-1)/2}, \qquad (\kappa_n)_{n \geq 0} = (1, 1, 0, -1, -1, 0, 1, 1, 0, \ldots). \tag{4}$$

The period is six. These two evaluations are credited prior inputs, not new assertions of this note. We work first in the field $\mathbb{Q}(q)$, where each $D_n^{(0)}$ is nonzero.

The Desnanot–Jacobi identity, applied to the $(n+1)$-square unshifted Hankel matrix, gives for every $n \geq 1$

$$D_{n+1}^{(0)} D_{n-1}^{(2)} = D_n^{(0)} D_n^{(2)} - (D_n^{(1)})^2. \tag{5}$$

Indeed, the two principal corner minors have shifts zero and two; each off-diagonal corner minor has shift one; and deleting both extreme rows and columns gives the size $n-1$, shift-two minor. This also applies to $n = 1$, with the empty determinant equal to one.

Set $E_n = D_n^{(2)} / D_{n+1}^{(0)}$. Dividing (5) by $D_n^{(0)} D_{n+1}^{(0)}$ and using $S_{n+1} - S_n = n^2$, we obtain

$$E_n - E_{n-1} = \frac{(D_n^{(1)})^2}{D_n^{(0)} D_{n+1}^{(0)}} = \kappa_n^2 q^{\,n(n-1) - n^2} = \kappa_n^2 q^{-n}.$$

Now $E_0 = 1 = \kappa_0^2$, and $\kappa_j^2 = 1$ precisely when $j \not\equiv 2 \pmod 3$. Summing the recurrence yields

$$E_n = \sum_{\substack{0 \leq j \leq n \\ j \not\equiv 2 \pmod 3}} q^{-j}.$$

Multiplying by $D_{n+1}^{(0)} = q^{S_n + n^2}$ proves (1). No division by a one-shift determinant occurs; its periodic zeros therefore cause no exception.

To normalize the result, the largest permitted index is n when $n \not\equiv 2 \pmod 3$, and $n-1$ otherwise. Reversing the summation about this largest index gives (2)–(3). The exponent c_n is nonnegative because $S_n \geq 0$ and $n^2 - n \geq 0$. Both sides of the evaluated identity are polynomials, so the equality obtained in $\mathbb{Q}(q)$ is also valid at $q = 0$ and at every complex q. This proves the theorem. □

The printed formula, scope, and checks

For example,

$$D_1^{(2)} = 1 + q, \qquad D_2^{(2)} = q^4(1 + q), \qquad D_3^{(2)} = q^{11}(1 + q^2 + q^3).$$

Thus the printed $f_3 = q^2 + q^3$ cannot be repaired by changing only its monomial prefactor: it has two nonzero terms, while $D_3^{(2)}$ has three. For every positive multiple of three one must add the constant term to f_n. The theorem supplies the full corrected evaluation for all sizes; a transcription correction alone would not qualify as a new campaign solution.

The accompanying verifier builds the moments directly from their defining recurrence and computes exact polynomial determinants. It checks all three shifts at sizes $0 \le n \le 16$, including zeros in the one-shift sequence, and compares the two forms of the answer. These finite checks audit the indexing and exponents. The proof for unbounded n is the condensation argument above.

The 2019 final article [3] was identified bibliographically but could not be obtained in full. The current preprint was inspected selectively, especially Appendix A; it was not read throughout. Consequently the source and priority audit is incomplete, and this package has zero new campaign credit. An authoring-agent audit and exact checks do not constitute independent peer review.

Task 2: a direct weighted-path interpretation

Construct a weight-preserving description of the terms $q^{S_n+n^2-j}$, $j \not\equiv 2 \pmod 3$, directly from the shifted Hankel path families. The objective is an explicit cancellation map whose fixed points explain the missing residue class.

Task 3: the three-shift determinant

Evaluate $D_n^{(3)}$ for every size and determine its sign and normalized factors in all three residue classes. Start from condensation with shift one; handle the vanishing $D_n^{(1)}$ without dividing by zero. This is a proposed direction, not a solved additional problem.

Task 4: higher even shifts

Prove or disprove coefficientwise positivity of $D_n^{(2k)}$ for all positive n, k, the separate Conjecture A.4 in [2]. The present theorem covers only $k = 1$. Positivity of a Gram determinant for real $q > 0$ does not establish coefficientwise positivity.

Task 5: retain a horizontal-step parameter

Replace the moment recursion by

$$M_{n+1}(a, q) = aM_n(a, q) + \sum_{r=0}^{n-1} q^{r+1} M_r(a, q) M_{n-r-1}(a, q).$$

Find the corresponding shift-two determinant and identify values of a giving periodic normalized one-shift determinants. Determine which specializations retain nonnegative coefficients in q.

References

[1] J. Cigler, *q-Catalan- und q-Motzkinzahlen*, Sitzungsber. Abt. II **208** (1999), 3–20, especially Eqs. (37), (39), (40), printed pp. 15–16. https://homepage.univie.ac.at/johann.cigler/preprints/cat-motzkin.pdf.

[2] S.-j. Oh and T. Scrimshaw, *Identities from representation theory*, arXiv:1805.00113v1 (2018), Appendix A, pp. 61–63. https://arxiv.org/abs/1805.00113v1.

[3] S.-j. Oh and T. Scrimshaw, *Identities from representation theory*, Discrete Math. **342**(9) (2019), 2493–2541. https://doi.org/10.1016/j.disc.2019.05.020. Bibliographic identification only; complete final text not inspected.